

Operational Technology (OT) Fundamentals and Industrial Control Systems

Operational Technology (OT) Fundamentals

Operational Technology (OT) refers to the hardware and software systems that monitor, control, and automate physical processes, devices, and industrial operations.

Core Functions

- Monitor physical processes in real time
- Control industrial equipment and machinery
- Ensure safe and reliable system operation

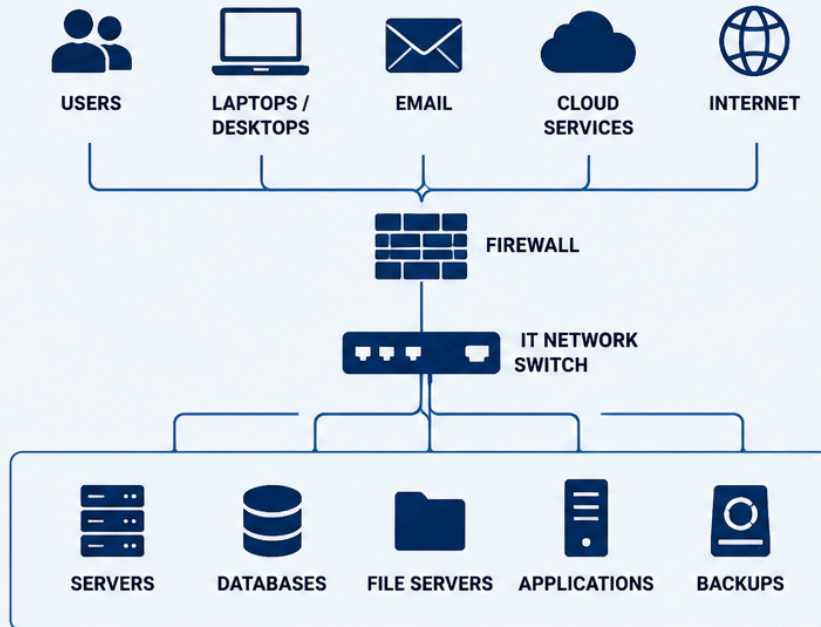


Information Technology vs Operational Technology

Information Technology (IT)	Operational Technology (OT)
Protects information	Controls physical processes
Confidentiality	Safety & Availability
Office environments	Industrial environments
Frequent software updates	Limited maintenance windows
Downtime acceptable	Downtime impacts production
Business protocols	Industrial protocols
Data-centric	Process-centric

INFORMATION TECHNOLOGY (IT)

Focus: Data & Business



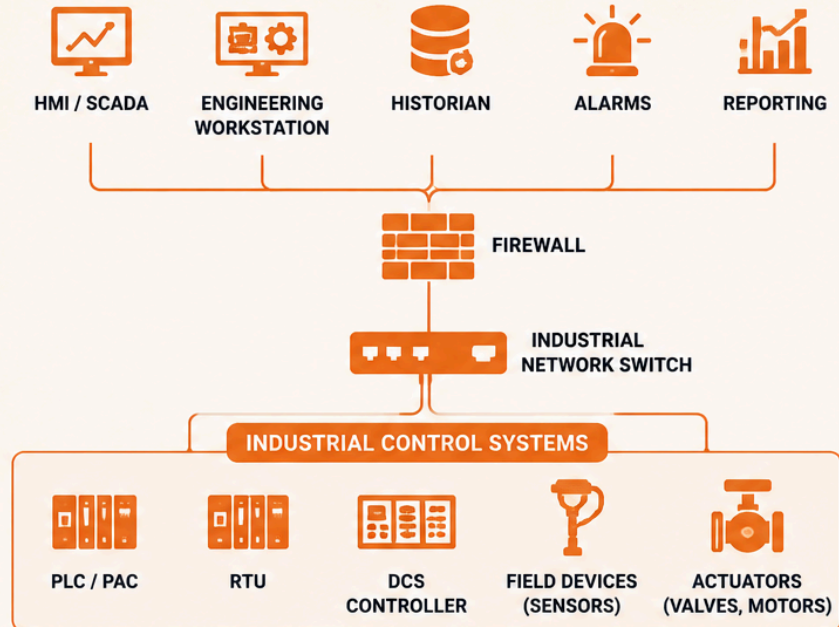
KEY CHARACTERISTICS

- Confidentiality is the top priority
- Standard protocols (TCP/IP, HTTP, HTTPS, SMTP, etc.)
- Frequent updates and patches
- Downtime is tolerable
- Business productivity is the primary concern

VS.

OPERATIONAL TECHNOLOGY (OT)

Focus: Operations & Processes



KEY CHARACTERISTICS

- Availability and Safety are the top priorities
- Industrial protocols (Modbus, DNP3, PROFINET, EtherNet/IP, etc.)
- Infrequent updates; long maintenance windows
- Downtime can halt production or cause safety incidents
- Process reliability and safety are the primary concern



IT PROTECTS INFORMATION. OT PROTECTS OPERATIONS.

Key Characteristics of Operational Technology

OT environments are engineered for **continuous, predictable, and reliable operation**.

Characteristics

- **Real-Time Operation** — Immediate response to process events
- **High Availability** — Continuous 24×7 operation
- **Deterministic Communication** — Predictable timing and behavior
- **Long Equipment Lifecycle** — Systems operate for decades
- **Safety-Critical Environment** — Failures may impact people, equipment, or the environment

Operational Technology is deployed across multiple critical infrastructure sectors.

• Common Industries •



Manufacturing



Power & Energy



Oil & Gas



Water & Wastewater



Transportation



Mining



Chemical Processing



Pharmaceuticals

Challenges in Operational Technology Security

OT cybersecurity differs significantly from traditional IT security due to operational and safety requirements.

Operational Challenges

- Legacy systems with limited security capabilities
- Continuous operation with minimal downtime
- Proprietary industrial communication protocols
- Limited patching opportunities
- Third-party vendor dependencies

Cybersecurity Challenges

- Flat industrial networks
- Lack of authentication in legacy protocols
- Increased remote access
- IT–OT convergence
- Expanding IIoT deployments

Industrial Control Systems (ICS)

Industrial Control Systems (ICS) are integrated hardware and software systems used to **monitor, control, and automate industrial processes** while ensuring **Safety, Reliability, Availability, and Process Integrity**.

Key Characteristics

Real-time operation | High availability | Continuous process control | Safety-critical environment

Core Components

Field Devices | PLC / RTU | Human Machine Interface (HMI) | SCADA | Historian |
Engineering Workstation | Industrial Network

Field Devices

Sensors

Convert physical parameters into electrical signals for controllers.

Measure:

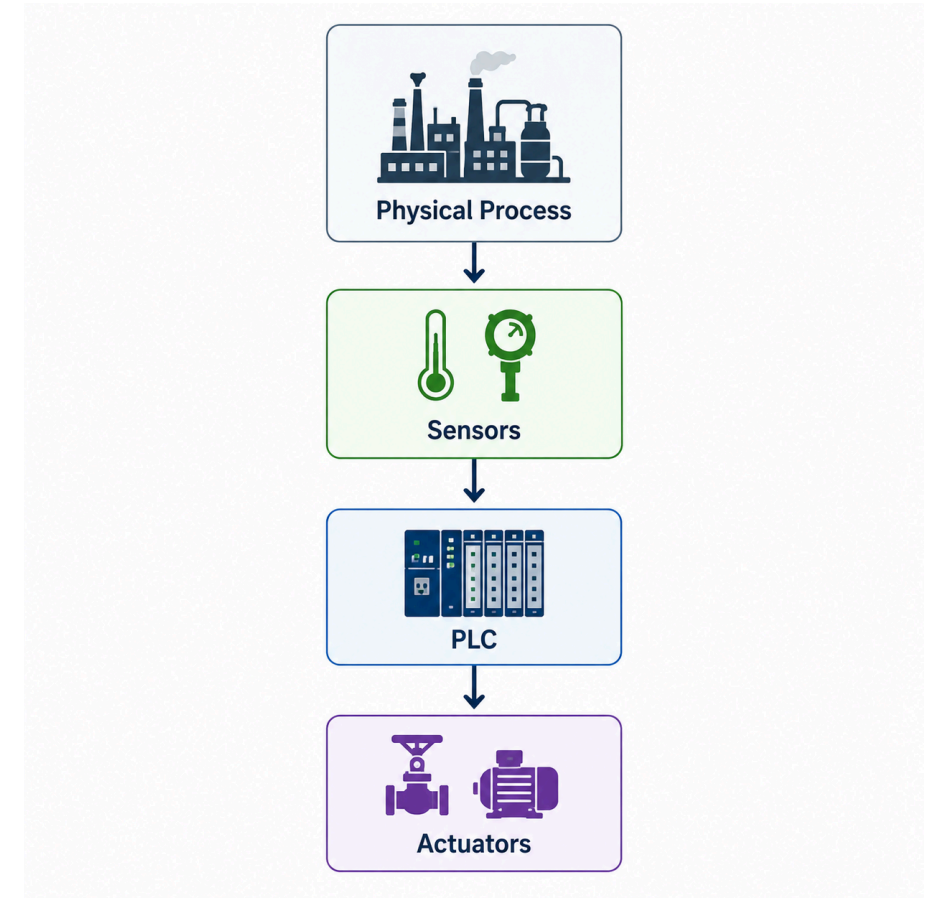
Temperature | Pressure | Flow | Level

Actuators

Receive commands from controllers and perform physical actions.

Open / Close Valves | Start / Stop Motors |

Control Pumps | Operate Relay



Programmable Logic Controller (PLC)

A Programmable Logic Controller (PLC) is an industrial computer designed to execute deterministic control logic and automate industrial processes.

Primary Functions

Read sensor inputs | Execute ladder logic/program | Make control decisions | Drive actuators

PLC Characteristics

High reliability and Continuous operation.

Supports industrial communication protocols.

The PLC is often considered the "brain" of an Industrial Control System.

Human Machine Interface (HMI)

Operator Interface for Industrial Processes

The **Human Machine Interface (HMI)** enables operators to monitor, supervise, and control industrial equipment through graphical dashboards.

Functions

- View process status
- Monitor alarms
- Display trends
- Start / Stop equipment
- Manual process control

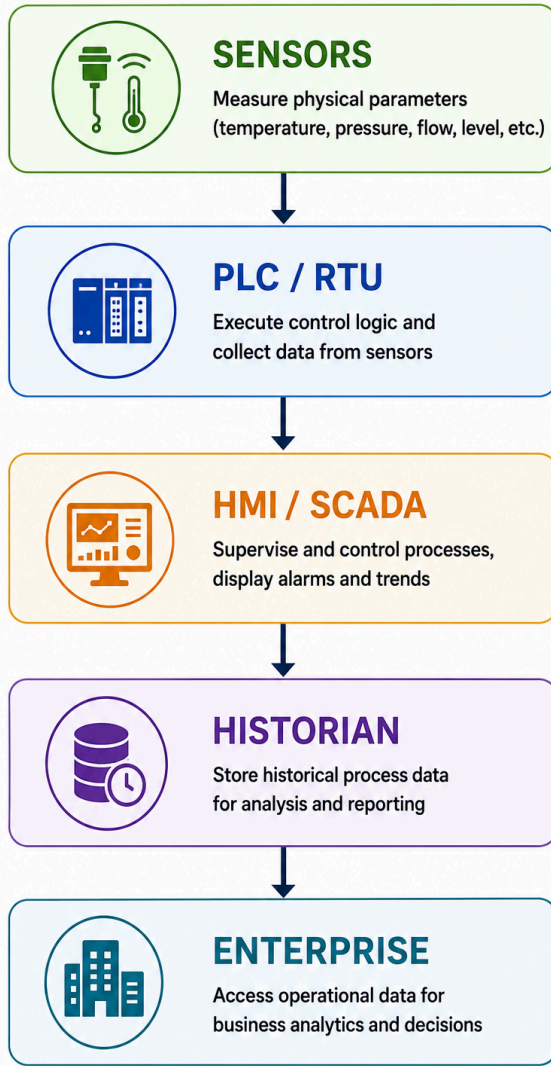
Supervisory Control and Data Acquisition (SCADA)

SCADA provides **centralized supervision and monitoring** of industrial assets distributed across large geographic areas.

Functions

- Collect field data
- Display process information
- Generate alarms
- Store operational data
- Remote monitoring
- Remote control

ICS Communication Flow



Purdue Enterprise Reference Architecture

The **Purdue Enterprise Reference Architecture (PERA)** is the most widely adopted framework for organizing Industrial Control Systems into logical layers.

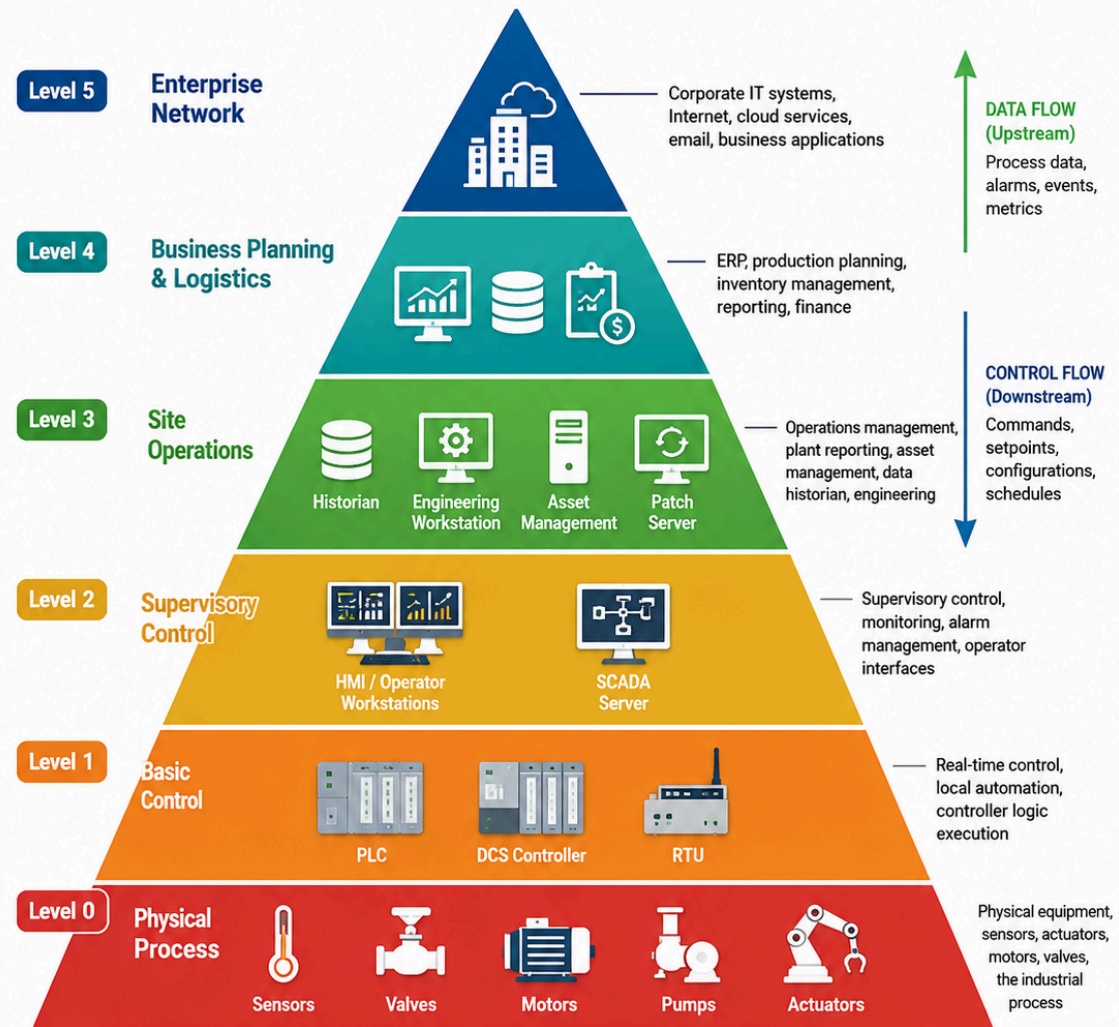
It separates operational functions based on **process control, network communication, and cybersecurity responsibilities**, reducing cyber risk while maintaining operational reliability.

Why Purdue Matters

- Defines logical operational zones
- Separates IT and OT environments
- Enables defense-in-depth
- Reduces attack propagation

Purdue Enterprise Reference Architecture

A hierarchical model for industrial automation and control systems



Key Principles



Segmentation
Separate levels and zones



Defense-in-Depth
Multiple layers of protection



Least Privilege
Access only what is necessary



Controlled Communication
Limit connections between levels



Monitoring & Visibility
Continuous monitoring across all levels

Level 0 — Physical Process

Where Industrial Operations Begin

Level 0 consists of the **physical equipment** that performs industrial operations.

Components

- Sensors | Actuators | Motors | Pumps | Valves | Turbines

Purpose

- Measure physical conditions
- Execute physical actions
- Interface directly with industrial processes

Level 1 — Basic Control

Local Process Control

Level 1 contains controllers responsible for executing deterministic control logic.

Components

- PLCs | RTU | Protection Relays | Embedded Controllers

Functions

- Read sensor values
- Execute control logic
- Control field devices
- Maintain continuous operation

Level 2 — Supervisory Control

Level 2 provides centralized monitoring and supervisory control over industrial processes.

Components

- HMI | SCADA Servers | Engineering Workstations | Alarm Systems

Functions

- Process visualization
- Alarm management
- Operator interaction
- Supervisory control

Level 3 — Manufacturing Operations

Level 3 coordinates production activities and operational management.

Components

- Historian | Manufacturing Execution System (MES) | Batch Servers | Application Servers | Patch Management | Backup Servers

Functions

- Production management
- Data collection
- Reporting
- Operational coordination

Level 4 — Business Network

Level 4 represents traditional enterprise IT systems supporting business operations.

Components

- ERP | Email Servers | Active Directory | Business Applications | Database Servers | Corporate File Servers

Functions

- Business planning
- Finance
- Inventory
- Human Resources
- Corporate IT services

Level 5 — Enterprise / Cloud

Level 5 extends enterprise operations beyond the organization.

Includes

- Cloud Services | Internet Connectivity | Remote Users | Vendor Connectivity | Corporate WAN

Primary Objective

Enable secure communication between industrial operations and external business services.

Industrial Communication Protocols

Enabling Reliable Communication Across Industrial Control Systems

Industrial communication protocols define how field devices, controllers, HMIs, SCADA systems, and enterprise applications exchange process data.

OT protocols prioritize:

- Deterministic communication
- Low latency
- High availability
- Reliability
- Real-time control

Core Functions

- Data Acquisition
- Process Monitoring
- Command Execution
- Alarm & Event Reporting
- System Diagnostics

Common Industrial Protocols

Protocol	Primary Use	Typical Industry
Modbus TCP	PLC Communication	Manufacturing
DNP3	Remote Telemetry	Power Systems
OPC UA	Secure Data Exchange	Industry 4.0
EtherNet/IP	Industrial Ethernet	Manufacturing
PROFINET	Real-Time Automation	Process Industry
IEC 60870-5-104	Power Automation	Utilities

Each protocol is designed to address specific industrial communication requirements.

Modbus TCP

One of the Most Widely Used Industrial Protocols

Features

- Master–Slave Architecture
- Client–Server (Modbus TCP)
- Open Standard
- Lightweight Communication

Security Limitations

- No Authentication
- No Encryption
- No Integrity Verification
- Susceptible to Command Injection
- Vulnerable to Replay Attacks

DNP3

Distributed Network Protocol Version 3

DNP3 is primarily used for communication between remote field devices and control centers within critical infrastructure.

DNP3 Features

- Data Acquisition
- Event Reporting
- Time Synchronization
- Remote Monitoring
- Control Operations

Advantages

- Efficient Event-Based Communication
- Designed for Wide Area Networks
- Supports Secure Authentication

DNP3 Security Considerations

- No Encryption (Legacy)
- Unauthorized Commands
- Traffic Manipulation
- Replay Attacks
- Critical Infrastructure Impact

OT Threat Landscape

Understanding Modern Cyber Threats to Industrial Control Systems

The **OT Threat Landscape** encompasses cyber threats, vulnerabilities, and attack techniques targeting Industrial Control Systems (ICS) and critical infrastructure.

Key Drivers

- IT–OT Convergence
- Industrial Internet of Things (IIoT)
- Remote Access
- Legacy Systems
- Supply Chain Dependencies

THE ICS CYBER KILL CHAIN

TYPICAL ATTACK PROGRESSION



OT Attack Consequences

Primary Objectives

- Ensure Safety
- Maintain Availability
- Preserve Reliability
- Protect Process Integrity
- Detect Malicious Activity
- Enable Rapid Recovery



IEC 62443

The International Standard for Industrial Cybersecurity

IEC 62443 is the globally recognized cybersecurity standard developed to secure **Industrial Automation and Control Systems (IACS)** throughout their entire lifecycle.

Objectives

- Protect Industrial Control Systems
- Reduce Cyber Risk
- Improve Operational Resilience
- Secure Critical Infrastructure
- Enable Lifecycle Security

- Personnel Safety
- Industrial Operations
- Equipment Availability
- Product Quality
- Environmental Protection
- Business Continuity

Why IEC 62443 Matters

Unlike traditional IT standards, IEC 62443 is specifically developed for **Operational Technology** environments where cybersecurity failures can impact physical processes, safety, and business continuity.



ADDRESSES



Industrial Control Systems



Manufacturing Plants



Utilities



Energy



Water



Oil & Gas



Transportation

The IEC 62443 Family

IEC 62443 consists of multiple standards grouped according to their intended audience and purpose.



Stakeholders in IEC 62443

Stakeholder	Primary Responsibility
Asset Owner	Operate and secure industrial systems
Product Supplier	Develop secure industrial products
System Integrator	Design and deploy secure systems
Service Provider	Maintain and support industrial systems

Defense-in-Depth in IEC 62443

IEC 62443 promotes a **layered security approach**, where multiple technical, physical, and administrative controls work together.

Security Layers

- Policies & Procedures
- Physical Security
- Network Security
- System Security
- Application Security



Zones and Conduits

Network Segmentation Based on Risk

IEC 62443 introduces the concepts of **Zones** and **Conduits** to improve network segmentation.

Zone

A logical or physical grouping of assets with similar cybersecurity requirements.

Conduit

A controlled communication path that securely connects different security zones.

Security Levels (SL)

IEC 62443 defines Security Levels (SL) based on the capability of potential attackers.

Security Level	Protection Against
SL1	Casual or accidental threats
SL2	Intentional attacks using simple means
SL3	Sophisticated attackers with moderate resources
SL4	Highly sophisticated attackers with extensive resources

IEC 62443 RISK ASSESSMENT

Risk assessment is the foundation of IEC 62443.

RISK ASSESSMENT PROCESS



1. Identify Assets



2. Identify Threats



3. Identify Vulnerabilities



4. Evaluate Consequences



5. Estimate Likelihood



6. Determine Risk



7. Select Security Controls



8. Residual Risk

Cyber Security Management System (CSMS)

IEC 62443-2-1

A Cyber Security Management System (CSMS) establishes governance, policies, procedures, and continuous improvement for industrial cybersecurity.

Core Elements

- Governance
- Risk Management
- Security Policies
- Incident Response
- Personnel Awareness
- Continuous Improvement

Patch Management

IEC 62443-2-3

Patch management reduces cyber risk while maintaining operational reliability.

IEC 62443-4-1

Product suppliers should integrate cybersecurity throughout the entire product development lifecycle.

Secure Development Practices

- Security Requirements
- Secure Design
- Secure Coding

Component Security

IEC 62443-4-2

Defines cybersecurity requirements for industrial devices such as:

- PLCs | RTU | HMI | Industrial Switches | Firewalls | Engineering Workstations

Foundational Requirements

- Identification & Authentication
- System Integrity and Data Confidentiality
- Restricted Data Flow
- Timely Response to Events
- Resource Availability

Key Takeaways

IEC 62443 provides a comprehensive framework for securing Industrial Automation and Control Systems.

- International cybersecurity standard for OT
- Risk-based approach
- Defense-in-Depth
- Lifecycle security
- Zones & Conduits
- Security Levels
- CSMS
- Patch Management
- Secure Development

Modbus TCP Frame Structure

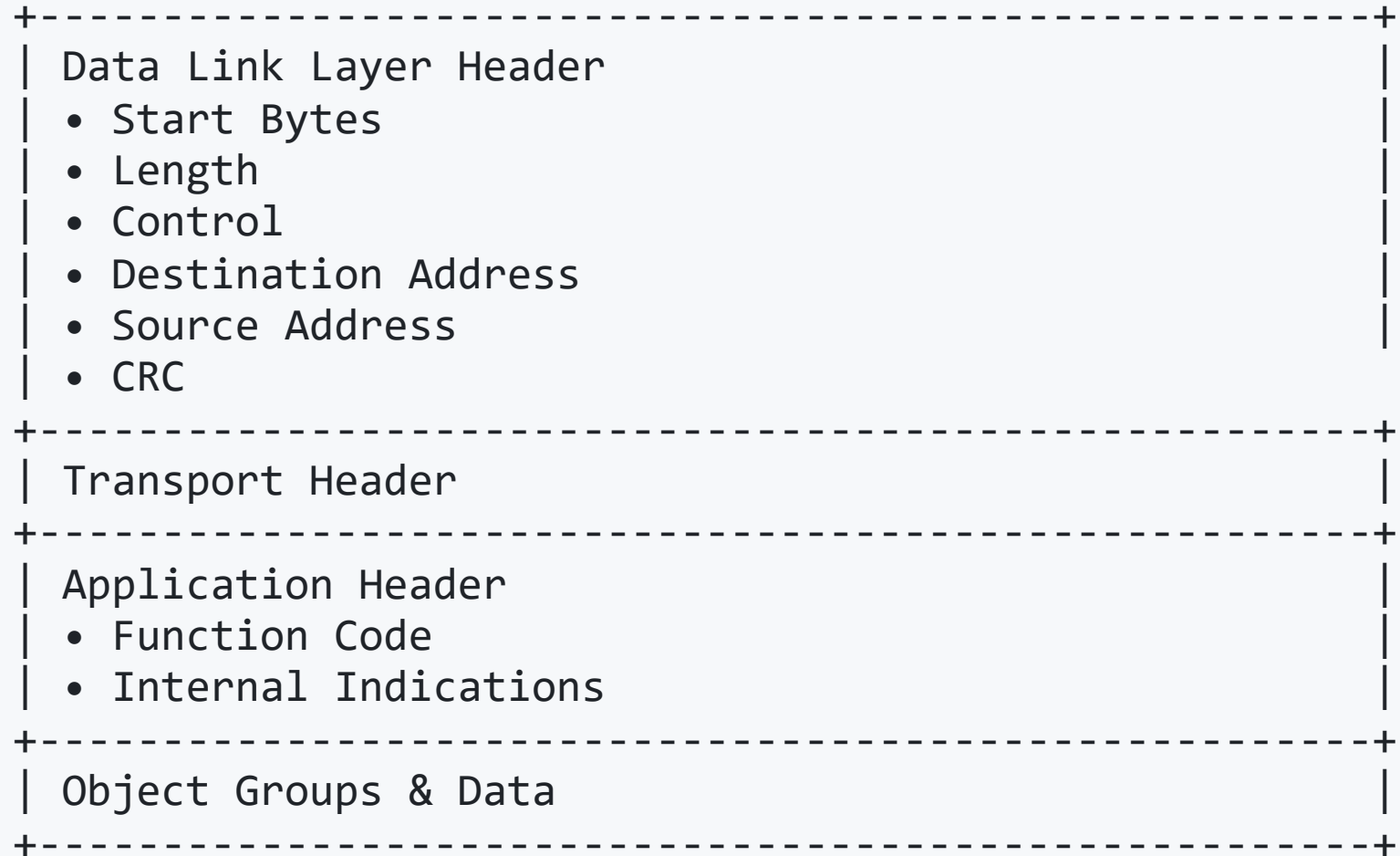
Ethernet Header
IPv4 Header
TCP Header (Port 502)
MBAP Header • Transaction ID • Protocol ID • Length • Unit Identifier
Protocol Data Unit (PDU) • Function Code • Data

Common Modbus Function Codes

Function Code	Operation	Category
FC01	Read Coils	Monitoring
FC03	Read Holding Registers	Monitoring
FC05	Write Single Coil	Control
FC06	Write Single Register	Control
FC15	Write Multiple Coils	Control
FC16	Write Multiple Registers	Control

- **Read Operations** retrieve process information.
- **Write Operations** modify the state of industrial devices.

DNP3 Frame Structure



Modbus TCP vs DNP3

Feature	Modbus TCP	DNP3
Communication	Client–Server	Master–Outstation
Default Port	TCP 502	TCP/UDP 20000
Packet Structure	MBAP Header + PDU	Data Link + Transport + Application Layers
Data Exchange	Polling-based	Event-driven + Polling
Function	Read/Write Coils & Registers	Read, Operate, Control, Event Reporting
Security	No Native Authentication or Encryption	Secure Authentication (DNP3-SA) Available

FPGA Data Diode Architecture

DATA DIODE ARCHITECTURE

